

A Harmonic-Near-Symmetric Order Obstruction at the Critical Sobolev Index

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Source Question and Status

Ivanov and Kalton study when a Riesz basis of exponentials in $L^2(-\pi, \pi)$ remains a basis in Sobolev spaces. After proving their unconditional-basis classification, they ask whether the exponentials can be a *conditional* basis of H^s , under a suitable ordering, throughout the intermediate range $s_0 \leq s \leq s_\Lambda$ [1, p. 14].

From the other hand, $\|e_{\lambda_n}\|_{H^s} \geq |\phi_n(e_{\lambda_n})|/\|\phi_n\|_{H^s}$, what gives $\|e_{\lambda_n}\|_{H^s} \geq C_1^{-1}(v_n^{1-s}h_n^s)^{1/2}$. Therefore the norms $\|e_{\lambda_n}\|_{H^s}$ and $\|e_{\lambda_n}\|_{Y_{1-s}}$ are both equivalent to $\sqrt{q_n}$. Therefore the assumption that (e_{λ_n}) is an unconditional basic sequence leads to equivalence of metrics H^s and Y_{1-s} . $\square$

Remark. It is easy to have necessary and sufficient condition for an exponential family $(e_{\lambda_n})_{n \in \mathbb{Z}}$ which is complete and minimal in L^2 to be complete and/or minimal in H^s [3]. To do this we connect the generating function F with the critical exponent s_Λ

$$s_\Lambda := \inf\{s : \int_{-\infty}^{\infty} \frac{|F(x)|^2}{1+|x|^{2s}} dx < \infty = \inf\{s : \psi \in (H^s)^*\}.$$

Now $(e^{i\lambda_n t})$ is complete in $H^s(-\pi, \pi)$ for $s < s_\Lambda$ and is minimal for $s > s_\Lambda - 1$. The situation for $s = s_\Lambda$ or $s = s_\Lambda - 1$ depends on whether ψ is bounded in H^{s_Λ} . Note that $s_0 \leq s_\Lambda \leq s_1$ in general.

Thus, the family (e_{λ_n}) is minimal in H^s for $0 < s < 1$, and for any $(\alpha_n) \in l^2$, $\alpha \neq 0$,

$$0 < \|\sum \alpha_n e_{\lambda_n} / \sqrt{q_n}\|_{H^s}^2 \leq \|\sum \alpha_n e_{\lambda_n} / \sqrt{q_n}\|_{Y_{1-s}}^2 \leq C \sum |\alpha_n|^2.$$

We do not know whether $(e_{\lambda_n})_{n \in \mathbb{Z}}$ can be a conditional basis of H^s , for some appropriate ordering, when $s_0 \leq s \leq s_\Lambda$.

In order to get precise estimates of s_0 and s_1 we will need to establish an alternative formula for $\|\psi\|_t$ in this special case.

Let us introduce the function $\Phi(z)$ defined by $\Phi(z) = |F(z)|d(z, \Lambda)^{-1}$ when $z \notin \Lambda$ and $\Phi(\lambda_n) = |F'(\lambda_n)|$ for $n \in \mathbb{Z}$. The function Φ plays an important role in the known conditions for (e_{λ_n}) to be an unconditional basis (see [20] and [19]). We will call Φ the *carrier function* for (e_{λ_n}) .

The following lemma lists some useful properties

This packet gives a rigorous negative answer for a broad class of orders in the canonical endpoint model $\Lambda = \mathbb{Z}$, where $s_0 = s_\Lambda = 1/2$. The full question for arbitrary, highly nonlocal permutations remains open.

Definitions

Write $I = (-\pi, \pi)$. We use the equivalent Slobodeckij norm

$$\|g\|_{H^{1/2}(I)}^2 \asymp \|g\|_{L^2(I)}^2 + [g]_{1/2,I}^2, \quad [g]_{1/2,I}^2 = \int_I \int_I \frac{|g(x) - g(y)|^2}{|x - y|^2} dx dy.$$

Let $\pi_0 : \mathbb{N} \rightarrow \mathbb{Z}$ be an enumeration and $A_k = \{\pi_0(1), \dots, \pi_0(k)\}$. We call the order *harmonic-near-symmetric along a subsequence* if there are $k_j, N_j \rightarrow \infty$ such that

$$\sum_{\substack{n \in A_{k_j} \triangle [-N_j, N_j] \\ n \neq 0}} \frac{1}{|n|} \rightarrow 0. \quad (1)$$

Proof Intuition

The smooth interval function $f(x) = x$ has Fourier coefficients of size $1/|n|$. Its symmetric Fourier sums converge in every subcritical $H^s(I)$, but at $s = 1/2$ their periodic jump is resolved in a boundary layer of width $1/N$. After rescaling that layer, the error tends to a nonconstant profile. The scale invariance of the critical Slobodeckij seminorm turns the profile into a fixed positive lower bound. Changing the retained frequencies by a set of harmonic mass $o(1)$ changes the sum by $o(1)$ in $H^{1/2}(I)$, so every order satisfying (1) inherits the defect.

Main Result

Theorem 1. *If an enumeration $\pi_0 : \mathbb{N} \rightarrow \mathbb{Z}$ is harmonic-near-symmetric along a subsequence, then the ordered system*

$$(e^{i\pi_0(k)x})_{k \geq 1}$$

is not a Schauder basis of $H^{1/2}(-\pi, \pi)$.

Proof. Let $f(x) = x$. This function belongs to $H^{1/2}(I)$, and with the normalization $\hat{f}(n) = (2\pi)^{-1} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$ one has

$$\hat{f}(0) = 0, \quad \hat{f}(n) = \frac{i(-1)^n}{n} \quad (n \neq 0).$$

Its symmetric Fourier sum is therefore

$$S_N f(x) = 2 \sum_{n=1}^N \frac{(-1)^{n+1} \sin(nx)}{n}.$$

We first prove that these sums do not converge to f in $H^{1/2}(I)$. At the boundary scale $x = \pi - u/N$ their errors satisfy

$$\begin{aligned} r_N(\pi - u/N) &:= f(\pi - u/N) - S_N f(\pi - u/N) \\ &= \pi - \frac{u}{N} - 2 \sum_{n=1}^N \frac{\sin(nu/N)}{n}. \end{aligned}$$

For u in any fixed compact subset of $[0, \infty)$,

$$\sum_{n=1}^N \frac{\sin(nu/N)}{n} = \frac{1}{N} \sum_{n=1}^N h(u, n/N), \quad h(u, t) = \begin{cases} \sin(ut)/t, & t > 0, \\ u, & t = 0, \end{cases}$$

is a Riemann sum converging uniformly to $\int_0^1 \sin(ut)/t \, dt$. Consequently

$$r_N(\pi - u/N) \longrightarrow R(u) := \pi - 2 \int_0^1 \frac{\sin(ut)}{t} \, dt \quad (2)$$

uniformly on compact u -intervals.

The profile is nonconstant: $R'(u) = -2\sin(u)/u$ for $u > 0$. Choose two disjoint compact intervals $U, V \subset (0, \pi)$ on which the values of R are separated by some $\delta > 0$. Uniform convergence in (2) then gives, for all sufficiently large N ,

$$|r_N(\pi - u/N) - r_N(\pi - v/N)| \geq \delta/2 \quad (u \in U, v \in V).$$

Restricting the Slobodeckij integral to the corresponding two boundary strips and changing variables yields

$$\begin{aligned} [r_N]_{1/2, I}^2 &\geq \int_U \int_V \frac{|r_N(\pi - u/N) - r_N(\pi - v/N)|^2}{|u - v|^2} \, du \, dv \\ &\geq c_{U, V, \delta} > 0. \end{aligned}$$

Thus $\liminf_N \|f - S_N f\|_{H^{1/2}(I)} > 0$.

For a finite set $A \subset \mathbb{Z}$, let $T_A f = \sum_{n \in A} \hat{f}(n) e^{inx}$. Restriction from the periodic space $H^{1/2}(\mathbb{T})$ to $H^{1/2}(I)$ is continuous. Hence, with $D = A \triangle [-N, N]$,

$$\begin{aligned} \|(T_A - S_N)f\|_{H^{1/2}(I)}^2 &\leq C \|(T_A - S_N)f\|_{H^{1/2}(\mathbb{T})}^2 \\ &\leq C' \sum_{\substack{n \in D \\ n \neq 0}} (1 + n^2)^{1/2} |\hat{f}(n)|^2 \leq C'' \sum_{\substack{n \in D \\ n \neq 0}} \frac{1}{|n|}. \end{aligned} \quad (3)$$

By (1), $\|(T_{A_{k_j}} - S_{N_j})f\|_{H^{1/2}(I)} \rightarrow 0$. The boundary lower bound and the triangle inequality imply

$$\liminf_j \|f - T_{A_{k_j}} f\|_{H^{1/2}(I)} > 0.$$

Finally, if the ordered exponentials were a Schauder basis, its expansion coefficients for f would have to be $\hat{f}(n)$. Indeed, convergence in $H^{1/2}(I)$ implies convergence in $L^2(I)$, where the integer exponentials are an orthogonal basis and coefficients are unique. Its k th basis partial sum would therefore be $T_{A_k} f$, contradicting the display above. $\square$

Concrete Orders Ruled Out

Let $\eta(1) = 0$, $\eta(2r) = r$, and $\eta(2r+1) = -r$ be the canonical symmetric enumeration, and let $p(m)$ be the position occupied by $\eta(m)$ in another enumeration.

Corollary 1. *The conclusion of Theorem 1 holds if either*

1. $\sup_m |p(m) - m| < \infty$, or

2. $|p(m) - m| = o(m)$ as $m \rightarrow \infty$.

In particular, the symmetric order, every bounded positional perturbation, and every sublinear positional perturbation fail at $s = 1/2$.

Proof. Let $B_M = \{\eta(1), \dots, \eta(M)\}$ and let A_M be the first M frequencies in the new order. If $|p(m) - m| \leq C$, membership in A_M and B_M can differ only when $|m - M| \leq C$; hence their harmonic symmetric difference has mass $O(C/M)$. If $|p(m) - m| = o(m)$, fix $\varepsilon > 0$. For all sufficiently large m , $(1 - \varepsilon)m \leq p(m) \leq (1 + \varepsilon)m$. Membership can then differ only for

$$\frac{M}{1 + \varepsilon} < m < \frac{M}{1 - \varepsilon},$$

apart from finitely many indices, which agree once M is large. This window has $O(\varepsilon M)$ indices, and $|\eta(m)| \asymp M$ there, so its harmonic mass is $O(\varepsilon)$. Taking $M = 2N + 1$ gives $B_M = [-N, N]$; first let $N \rightarrow \infty$ and then $\varepsilon \downarrow 0$ to obtain (1). $\square$

Scope, Upgrade Attempts, and Novelty Check

The theorem is not a full answer to the Ivanov–Kalton question. An arbitrary permutation can retain a reservoir of remote frequencies with positive harmonic mass while delaying frequencies at every scale, so (3) need not compare any of its prefixes with a symmetric sum. The argument also does not cover general noninteger Λ .

Eight focused upgrade attempts were made before promotion: a false Bessel-to-Riesz implication was eliminated; uniform minimality was checked; the lattice endpoint was reduced to exact spectral finite sections; the boundary profile was proved; it was extended to harmonic-near-symmetric orders; arbitrary permutations were tested and isolated as the obstruction; weighted conditional exponential bases were examined as possible positive models; and regular split-shift examples from the source were stress-tested. The last computation again showed endpoint growth for natural symmetric orders but supplied no theorem for exotic permutations.

The run’s cheap indexes and a bounded exact-phrase/citation search through 17 August 2026 found no later answer to the exact question. This theorem should therefore be reviewed as a new ordering-sensitive partial result, not as a claim that the source problem is resolved.

Verification Notes

The proof is analytic. The companion script evaluates the limiting profile, checks finite- N convergence on a fixed grid, and verifies explicit separation between two profile intervals. These computations are consistency checks only and are not used in the proof.

Human Review Recommendation

Verify the critical boundary rescaling, the periodic-to-interval norm comparison in (3), and the positional-displacement corollary. Most importantly, preserve the distinction between the proved harmonic-near-symmetric obstruction and the unresolved arbitrary-order class.

References

- [1] S. Ivanov and N. Kalton, *Interpolation of subspaces and applications to exponential bases in Sobolev spaces*, arXiv:math/0104130; St. Petersburg Math. J. **13** (2002), no. 2, 221–239.